

Small Bowel, Malabsorption, Diarrheal Syndromes, and Bowel Obstruction

Medvale Internal Medicine - Gastroenterology and Hepatology

This chapter is an original teaching chapter built from coverage signals in the GI source packet and the broader Medvale internal medicine blueprint. The emphasis is on clinically useful illness scripts, diagnostic branching, numeric anchors, and entity relationships that can later be extracted into a graph RAG system.

Why this matters clinically

Small bowel disease sits at the intersection of emergency medicine, inpatient medicine, outpatient gastroenterology, nutrition, and immunology. A patient with small bowel obstruction can deteriorate within hours if strangulation or ischemia develops. A patient with malabsorption may present more slowly, with anemia, osteopenia, neuropathy, edema, bruising, or weight loss rather than an obvious bowel complaint. A patient with chronic diarrhea may have a benign functional syndrome, but the same symptom can be the first clue to inflammatory bowel disease, celiac disease, bile acid diarrhea, medication toxicity, endocrine disease, pancreatic insufficiency, or chronic infection.

For exam-style reasoning, the key is to divide the problem into a small number of clinical buckets: obstructive versus nonobstructive distention; acute versus chronic diarrhea; watery versus fatty versus inflammatory stool; mucosal disease versus pancreatic or bile-acid disease; and functional symptoms versus red flags. In practice, these categories overlap, but the first branch usually determines the test that prevents delay: CT abdomen for obstruction, stool studies for infectious or inflammatory diarrhea, celiac serology for malabsorption, and endoscopy/biopsy when structural disease is likely.

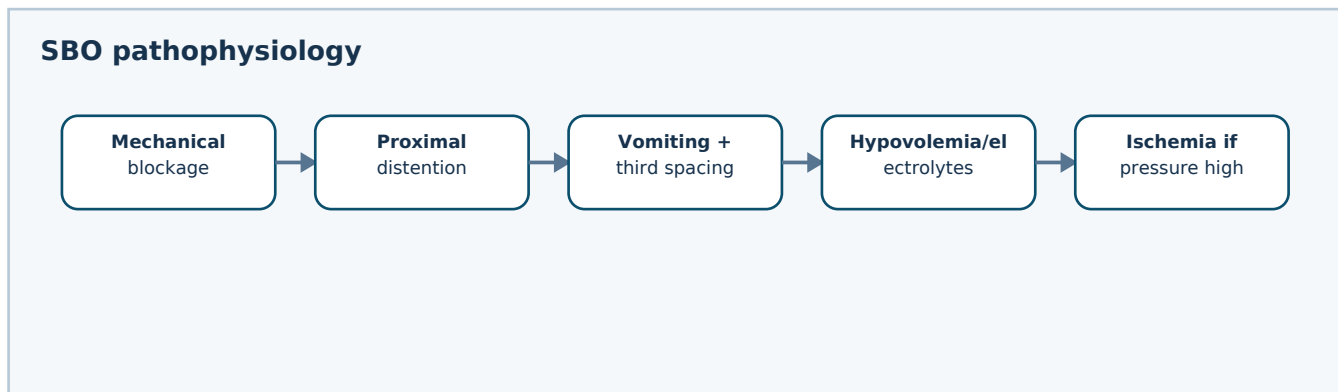
Core illness scripts

Problem	Classic pattern	Immediate clinical decision
Small bowel obstruction	Colicky abdominal pain, vomiting, distention, obstipation; prior surgery or hernia is common	Look for strangulation/ischemia; resuscitate, decompress, image, and involve surgery early
Paralytic ileus	Diffuse gas and bowel dilation without a transition point; often postoperative, medication-related, electrolyte-related, or inflammatory	Correct cause; avoid unnecessary laparotomy unless diagnosis changes
Celiac disease	Chronic diarrhea, bloating, iron deficiency, weight loss, dermatitis herpetiformis, osteoporosis, or abnormal liver enzymes	Test while patient eats gluten; confirm when needed; lifelong gluten-free diet
SIBO	Bloating, diarrhea, flatulence, malabsorption in a patient with dysmotility, altered anatomy, blind loop, or systemic sclerosis	Breath testing or empiric treatment in selected settings; treat cause when possible
Inflammatory diarrhea	Blood, fever, nocturnal symptoms, tenesmus, weight loss, high fecal calprotectin/lactoferrin	Rule out infection and evaluate for IBD, ischemia, or malignancy
Fatty diarrhea	Bulky, greasy, floating, foul stool with weight loss or fat-soluble vitamin deficiency	Localize to mucosal disease, pancreatic enzyme failure, bile acid deficiency, or lymphatic obstruction

Small bowel obstruction: mechanism and triage

Small bowel obstruction (SBO) is a mechanical blockage that prevents normal transit of gas, fluid, and intestinal contents. In adults, adhesions after prior abdominal or pelvic surgery are the most common cause; incarcerated hernias are another major cause because they can obstruct and strangulate a loop of bowel. Other causes include Crohn disease strictures, malignancy, carcinomatosis, intussusception, gallstone ileus, volvulus, radiation strictures, and superior mesenteric artery syndrome compressing the third portion of the duodenum.

The immediate danger is not simply that stool cannot pass. Distended bowel secretes fluid, loses absorptive capacity, and triggers vomiting. This creates intravascular volume depletion, hypochloremia, hypokalemia, and metabolic alkalosis. If intraluminal pressure rises high enough to compromise venous drainage and then arterial inflow, the bowel becomes ischemic. Ischemia can progress to necrosis, perforation, peritonitis, sepsis, and shock. Closed-loop obstruction is especially dangerous because the bowel is occluded at two points, so pressure rises rapidly within an isolated segment.



Numeric anchors and exam clues

A complete obstruction usually causes obstipation, but residual distal colonic gas or stool can make early passage of flatus misleading. Severe continuous pain, fever, tachycardia, leukocytosis, peritoneal signs, metabolic acidosis, elevated lactate, pneumatosis, portal venous gas, free air, or a closed-loop configuration should shift the clinician away from prolonged conservative therapy and toward urgent operative evaluation. The source packet emphasizes this same clinical danger: high intraluminal pressure may compromise blood supply and lead to strangulation, shock, gangrene, peritonitis, or perforation; it also lists fever, continuous severe pain, hematemesis, shock, gas in the bowel wall or portal vein, free air, peritoneal signs, and acidosis as concerning features for strangulated bowel.

Diagnosis of suspected obstruction

CT abdomen and pelvis with IV contrast is the practical modern test of choice for most stable adults with suspected SBO because it can identify the transition point, grade the obstruction, detect hernia or mass, and assess for ischemia. Plain abdominal radiographs may show dilated small bowel loops and air-fluid levels, but they are less sensitive and less specific than CT. A small bowel follow-through or water-soluble contrast challenge can help in selected adhesive SBO cases by clarifying partial obstruction and predicting nonoperative resolution. Barium is generally avoided when perforation is possible; water-soluble contrast is safer when leakage is a concern.

Feature	Small bowel obstruction	Large bowel obstruction	Paralytic ileus
Pain	Crampy, colicky; severe continuous pain suggests ischemia	Crampy with marked distention; cecal perforation risk if severe	Dull distention/discomfort; often less colicky
Vomiting	Earlier and more prominent with proximal SBO	Later unless ileocecal valve incompetent	Variable; often postoperative nausea
Distention	More prominent with distal obstruction	Often marked	Diffuse

Feature	Small bowel obstruction	Large bowel obstruction	Paralytic ileus
Gas pattern	Dilated small bowel proximal to transition; little distal gas if complete	Dilated colon proximal to obstruction	Gas distributed through small bowel and colon without fixed transition
Initial management	NPO, IV fluids, NG decompression if vomiting/distended, electrolytes, surgery consult	CT and surgical/GI evaluation; decompression strategy depends on cause	Treat cause: opioids/anticholinergics, hypokalemia, sepsis, peritonitis, postoperative state

Management begins before the final cause is known: make the patient NPO, establish IV access, give isotonic fluids, correct potassium and chloride deficits, monitor urine output, provide antiemetics and analgesia, and place a nasogastric tube when vomiting, severe distention, or aspiration risk is present. Antibiotics are used when strangulation, ischemia, perforation, fever, leukocytosis, or sepsis is suspected. Surgery is indicated for peritonitis, perforation, ischemia, closed-loop obstruction, complete obstruction with concerning course, incarcerated hernia, or failure of appropriate nonoperative management. The Bologna adhesive SBO guideline reports substantial morbidity, an average hospital stay around 8 days, and in-hospital mortality around 3% per episode, which explains why early risk stratification matters.

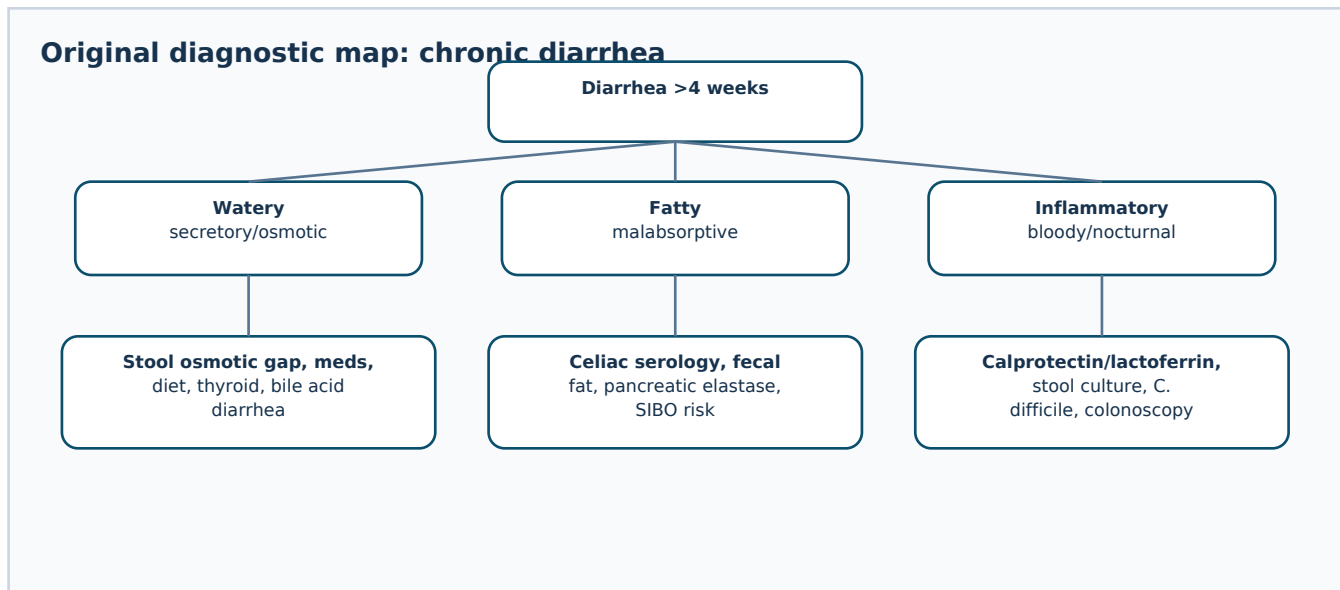
Paralytic ileus and pseudo-obstruction

Paralytic ileus is impaired propulsion without a mechanical transition point. It is common after abdominal surgery and is promoted by opioids, anticholinergic drugs, spinal cord injury, sepsis, shock, peritonitis, pancreatitis, and electrolyte disorders, especially hypokalemia. Imaging tends to show diffuse gas throughout the small bowel and colon rather than a single transition point. The treatment is supportive: bowel rest when needed, IV fluids, correction of potassium and magnesium abnormalities, minimizing opioids and anticholinergics, mobilization, and NG decompression only if vomiting or severe distention is present.

Acute colonic pseudo-obstruction, also called Ogilvie syndrome, is functional colonic dilation without a mechanical lesion. It occurs in hospitalized or postoperative patients, especially older adults or those with severe illness, trauma, electrolyte abnormalities, neurologic disease, or medications that suppress motility. The danger is cecal ischemia and perforation as diameter rises. Management includes correcting triggers, stopping offending medications, decompression when needed, and escalation to neostigmine or colonoscopic decompression in selected patients under appropriate monitoring.

Chronic diarrhea: first classify the stool

Chronic diarrhea is usually defined as loose or watery stools lasting more than 4 weeks. The first diagnostic move is to decide whether the stool is watery, fatty, or inflammatory. Watery diarrhea can be osmotic or secretory. Osmotic diarrhea improves with fasting and is caused by poorly absorbed solutes such as lactose, sorbitol, magnesium, or carbohydrate malabsorption. Secretory diarrhea persists during fasting and can be caused by bile acid diarrhea, microscopic colitis, endocrine tumors, some infections, laxatives, and medications. Fatty diarrhea suggests maldigestion or malabsorption. Inflammatory diarrhea suggests mucosal injury from IBD, infection, ischemia, radiation, or malignancy.



Category	Mechanism	Typical clues	Useful tests
Watery osmotic	Unabsorbed solute pulls water into lumen	Improves with fasting; bloating/gas; lactose, sorbitol, Mg, tube feeds	Diet/medication review; stool osmotic gap often >100 mOsm/kg
Watery secretory	Active secretion or impaired absorption of electrolytes/water	Persists with fasting; nocturnal diarrhea; large volume	Stool osmotic gap often <50 mOsm/kg; consider bile acid diarrhea, microscopic colitis, endocrine disease
Fatty	Impaired fat digestion or absorption	Bulky, greasy, floating, foul stool; weight loss; ADEK deficiency	Fecal fat, fecal elastase, celiac testing, pancreatic imaging if indicated
Inflammatory	Mucosal inflammation or ulceration	Blood, fever, tenesmus, nocturnal stool, anemia, weight loss	Stool culture, C. difficile, fecal calprotectin/lactoferrin, colonoscopy with biopsy

The AGA guideline for evaluation of chronic watery diarrhea supports using fecal calprotectin or fecal lactoferrin to screen for inflammatory bowel disease in patients with suspected functional diarrhea or IBS-D, rather than relying on ESR or CRP alone. A low-risk patient with long-standing symptoms, normal weight, no anemia, no blood, no nocturnal symptoms, and negative inflammatory markers is much more likely to have functional diarrhea or IBS-D. A patient with weight loss, fever, blood, iron deficiency, hypoalbuminemia, age over 50 at new onset, family history of colorectal cancer/IBD/celiac disease, or abnormal inflammatory stool markers needs a structural and inflammatory workup.

Malabsorption: localize the failed step

Malabsorption is not one disease; it is a failure in the sequence of nutrient delivery, digestion, mucosal uptake, lymphatic transport, or enterohepatic recycling. The same patient can have diarrhea plus anemia, neuropathy, bruising, edema, osteoporosis, infertility, or weight loss. The pattern of deficiency helps localize the lesion. Iron

and folate deficits point toward proximal small bowel disease; vitamin B12 and bile acid problems point toward terminal ileal disease; fat-soluble vitamin deficiency suggests fat malabsorption from mucosal disease, pancreatic insufficiency, cholestasis, bile acid deficiency, or lymphatic obstruction.

Failed step	Representative causes	Stool/clinical pattern	Deficiency associations
Mucosal absorption	Celiac disease, tropical sprue, Crohn disease, radiation enteritis	Diarrhea, bloating, weight loss; may have iron deficiency	Iron, folate, calcium, vitamin D; severe disease can affect ADEK
Luminal digestion	Chronic pancreatitis, pancreatic cancer, cystic fibrosis	Steatorrhea with weight loss; pancreatic pain/history may be present	ADEK, B12 sometimes; low fecal elastase
Bile acid delivery	Cholestasis, ileal resection, severe ileal Crohn	Fat malabsorption if bile salt pool depleted; watery diarrhea if bile acids spill into colon	ADEK; B12 if terminal ileum involved
Bacterial deconjugation/competition	SIBO from dysmotility, blind loop, systemic sclerosis, altered anatomy	Bloating, gas, diarrhea, sometimes steatorrhea	B12 deficiency; folate may be elevated
Lymphatic transport	Intestinal lymphangiectasia, lymphoma	Edema, protein-losing enteropathy, chylous fluid	Fat-soluble vitamins, lymphopenia, hypoalbuminemia

Celiac disease

Celiac disease is an immune-mediated enteropathy triggered by gluten-containing grains in genetically susceptible patients. Gluten peptides are modified by tissue transglutaminase and presented to T cells, driving villous injury, crypt hyperplasia, and intraepithelial lymphocytosis. The result is a reduced absorptive surface area, especially in the proximal small bowel, where iron, folate, and calcium absorption are important. The classic patient has chronic diarrhea, bloating, fatigue, weight loss, and abdominal distention, but many adults present with iron deficiency anemia, infertility, osteoporosis, aphthous ulcers, mild transaminase elevation, dermatitis herpetiformis, or family screening rather than dramatic diarrhea.

Important numeric anchors include an approximate 1% prevalence in many Western populations, an increased risk among first-degree relatives, and dermatitis herpetiformis in a minority of patients. The source packet highlights first-degree relative risk around 10% and dermatitis herpetiformis in 10% to 20% of patients, which are useful memory anchors. Testing should be performed while the patient is eating gluten. The usual first test is IgA tissue transglutaminase antibody with a total IgA level. If IgA deficient, use IgG-based testing such as deamidated gliadin peptide IgG or tTG-IgG. Duodenal biopsy confirms villous atrophy when diagnosis is uncertain, when serology and symptoms do not align, or when adult confirmation is needed before a lifelong restrictive diet.

Domain	High-yield associations
Symptoms	Diarrhea, bloating, flatulence, abdominal distention, fatigue, weight loss; symptoms may be absent
Deficiencies	Iron deficiency, folate deficiency, calcium/vitamin D deficiency, fat-soluble vitamin deficiency in severe disease
Extraintestinal	Dermatitis herpetiformis, aphthous ulcers, infertility, osteoporosis/osteopenia, neurologic symptoms, abnormal liver enzymes
Testing	tTG-IgA plus total IgA while on gluten; biopsy if needed; evaluate nutritional deficits
Treatment	Strict lifelong gluten-free diet; dietitian support; monitor symptoms, serology, and deficiencies
Pitfall	Do not start gluten-free diet before testing unless necessary; negative tests after gluten withdrawal can be misleading

Treatment is not simply symptom control. A strict gluten-free diet aims for mucosal healing, correction of nutritional deficiencies, prevention of bone disease, and reduction of long-term complications. Re-check symptoms and serologies, assess adherence, and replace iron, folate, B12, calcium, vitamin D, and fat-soluble vitamins when deficient. Persistent symptoms despite a gluten-free diet usually reflect ongoing gluten exposure, lactose intolerance, IBS overlap, microscopic colitis, SIBO, pancreatic insufficiency, or refractory celiac disease.

Small intestinal bacterial overgrowth

Small intestinal bacterial overgrowth (SIBO) occurs when excessive bacteria occupy the small intestine and interfere with digestion and absorption. Predisposing states include abnormal anatomy after surgery, blind loops, strictures, fistulas, ileocecal valve loss, motility disorders, systemic sclerosis, diabetic autonomic neuropathy, chronic opioid use, and sometimes chronic acid suppression in susceptible patients. Symptoms are nonspecific: bloating, abdominal discomfort, flatulence, diarrhea, and sometimes steatorrhea or weight loss. Laboratory clues can include B12 deficiency because bacteria consume cobalamin, and folate may be normal or elevated because bacteria produce folate.

Breath testing with glucose or lactulose is commonly used, but sensitivity and specificity are imperfect. ACG guidance defines SIBO as excessive bacteria in the small bowel causing gastrointestinal symptoms and reviews breath testing and treatment options. Rifaximin is commonly used for hydrogen-predominant SIBO, whereas methane-predominant constipation-associated overgrowth may require a different regimen. The durable intervention is treating the reason bacteria accumulated: improve motility when possible, address strictures or blind loops, reduce unnecessary opioids, and correct nutritional deficiencies.

Tropical sprue and Whipple disease

Tropical sprue is an acquired malabsorptive syndrome in people living in or returning from tropical regions. It can mimic celiac disease with chronic diarrhea, weight loss, fatigue, and folate or B12 deficiency. Biopsy can show villous blunting, but the epidemiologic setting and response to antibiotics plus folate distinguish it from celiac disease. Treatment is prolonged antibiotic therapy, often with folate repletion, and symptoms may take months to fully resolve.

Whipple disease is a rare systemic infection caused by *Tropheryma whipplei*. It classically causes weight loss, diarrhea, abdominal pain, migratory arthralgias or arthritis, and sometimes fever, lymphadenopathy, hyperpigmentation, neurologic findings, or culture-negative endocarditis. Diagnosis is supported by small bowel biopsy showing PAS-positive macrophages in the lamina propria and confirmed with organism-specific testing when available. Treatment requires prolonged antibiotics, often beginning with an induction phase that penetrates the central nervous system followed by long oral therapy, because relapse can be severe.

Lactose intolerance, bile acid diarrhea, and pancreatic insufficiency

Lactose intolerance is carbohydrate malabsorption from lactase deficiency. Undigested lactose reaches the colon, where bacterial fermentation produces gas and osmotic solutes. Symptoms include bloating, flatulence, cramps, and watery diarrhea after dairy intake. The condition is common and can be primary or secondary to mucosal injury such as viral gastroenteritis, celiac disease, Crohn disease, or chemotherapy. Diagnosis is often clinical through dietary elimination and rechallenge; hydrogen breath testing can support uncertain cases.

Bile acid diarrhea occurs when excess bile acids reach the colon and stimulate secretion. It can follow terminal ileal resection, ileal Crohn disease, cholecystectomy, or idiopathic bile acid malabsorption. Mild bile acid spillover causes watery diarrhea, while severe bile salt depletion can also impair micelle formation and cause fat malabsorption. Empiric therapy with a bile acid sequestrant may be used when suspicion is high, but clinicians should remember that sequestrants can worsen bloating and interfere with absorption of other medications.

Pancreatic exocrine insufficiency causes maldigestion because lipase and proteases do not reach the intestinal lumen in sufficient quantity. Chronic pancreatitis, pancreatic cancer, pancreatic surgery, cystic fibrosis, and severe longstanding diabetes can cause it. The classic stool is bulky, greasy, floating, and foul-smelling, but symptoms may be subtle until enzyme output is markedly reduced. Fecal elastase is a useful noninvasive test; treatment is pancreatic enzyme replacement with meals plus fat-soluble vitamin monitoring.

IBS, microscopic colitis, and IBD bridge

Irritable bowel syndrome is a disorder of gut-brain interaction characterized by recurrent abdominal pain related to defecation or associated with changes in stool frequency or form. IBS can cause diarrhea, constipation, or mixed bowel habits, but it should not cause GI bleeding, nocturnal diarrhea, persistent fever, progressive weight loss, iron deficiency anemia, hypoalbuminemia, or objective inflammation. Those features are red flags that require another diagnosis. Postinfectious IBS is common after bacterial gastroenteritis and can persist even after the pathogen is gone.

Microscopic colitis causes chronic watery nonbloody diarrhea, often in older adults, and the colon frequently looks normal during colonoscopy. Diagnosis requires random colon biopsies. Medication associations include NSAIDs, PPIs, SSRIs, and others. Smoking and autoimmune disease are risk factors. Treatment begins with stopping triggers and antidiarrheals in mild cases; budesonide is commonly used for more active disease.

Inflammatory bowel disease will receive its own colon-focused chapter, but it belongs in the chronic diarrhea differential. Crohn disease can affect any part of the GI tract and commonly involves the terminal ileum, producing abdominal pain, diarrhea, weight loss, strictures, fistulas, abscesses, and B12 or bile acid problems. Ulcerative colitis is limited to the colon and rectum and causes continuous mucosal inflammation with bloody diarrhea and urgency. Epidemiologic anchors from the source packet include onset often between ages 15 and 35, higher prevalence in Caucasian and Jewish populations, and important familial clustering.

Medication-induced and endocrine diarrhea

Medication review is a diagnostic test. Common causes of diarrhea include antibiotics, metformin, magnesium-containing antacids, colchicine, SSRIs, NSAIDs, PPIs, mycophenolate, orlistat, acarbose, chemotherapy, immunotherapy, laxatives, and alcohol. Endocrine and systemic causes include hyperthyroidism, adrenal insufficiency, diabetes-related autonomic dysfunction, carcinoid syndrome, VIPoma, gastrinoma, and mast cell disease. These diagnoses are uncommon compared with medications, infection, IBS, and IBD, but they become important when diarrhea is high-volume, nocturnal, fasting-persistent, associated with flushing or wheezing, or accompanied by major electrolyte abnormalities.

Diagnostic approach: what to order first

The initial workup should match the pretest probability. Acute severe diarrhea with fever, blood, dehydration, recent antibiotics, immunocompromise, outbreak exposure, or travel needs infectious testing. Chronic watery diarrhea without alarm features can begin with CBC, metabolic panel, thyroid testing when suggested, celiac serology, medication/diet review, and fecal calprotectin or lactoferrin to screen for inflammatory disease. Chronic fatty diarrhea needs fecal fat or elastase and evaluation for celiac, pancreatic, hepatobiliary, and ileal disease. Chronic inflammatory diarrhea needs stool culture or multiplex pathogen testing, *C. difficile* testing when appropriate, inflammatory markers, and colonoscopy with biopsy.

Clinical situation	Reasonable initial tests	Escalation trigger
SBO suspected	CBC, CMP, lactate if ill, CT abdomen/pelvis with IV contrast, pregnancy test when relevant	Peritonitis, closed-loop, ischemia, free air, strangulated hernia, persistent complete obstruction
Chronic watery diarrhea	Medication review, CBC/CMP, celiac serology, fecal calprotectin/lactoferrin, stool studies if exposure risk	Alarm features, nocturnal stool, abnormal markers, age/risk factors
Fatty diarrhea	Fecal fat or qualitative screen, fecal elastase, celiac testing, liver/pancreas imaging if indicated	Weight loss, hypoalbuminemia, vitamin deficiencies, pancreatic cancer concern
Possible celiac disease	tTG-IgA and total IgA while eating gluten; nutritional labs	Positive serology, IgA deficiency, high suspicion despite negative tests
Possible microscopic colitis	Colonoscopy with random biopsies even if mucosa appears normal	Persistent watery diarrhea in older adult or medication-associated syndrome
Possible IBD	Fecal calprotectin/lactoferrin, stool infection tests, colonoscopy with ileoscopy/biopsy	Blood, anemia, weight loss, nocturnal symptoms, fistula/abscess, high inflammatory markers

Management principles by syndrome

Syndrome	First-line management	Avoid/remember
Small bowel obstruction	NPO, IV fluids, electrolyte correction, NG decompression when needed, antiemetics/analgesia, surgical consultation	Do not delay surgery when ischemia/peritonitis/closed loop is suspected
Paralytic ileus	Treat precipitant, mobilize, reduce opioids/anticholinergics, correct K/Mg, supportive care	Surgery is usually not helpful unless obstruction/perforation is actually present
Celiac disease	Strict gluten-free diet, dietitian, replete deficiencies, monitor serology and symptoms	Testing is unreliable after gluten withdrawal without challenge
SIBO	Treat underlying anatomy/motility issue; antibiotics such as rifaximin in selected cases; nutritional repletion	Breath tests are imperfect; relapse is common if cause persists
Bile acid diarrhea	Bile acid sequestrant trial when suspicion high	Can worsen bloating and bind other drugs
Pancreatic insufficiency	Pancreatic enzyme replacement with meals, nutrition support, ADEK monitoring	Look for pancreatic cancer if new onset with weight loss or pain
IBS-D/functional diarrhea	Reassurance after red flags excluded; diet, soluble fiber, antidiarrheals, gut-brain therapies as appropriate	Do not label as IBS when bleeding, anemia, weight loss, nocturnal stool, or objective inflammation exists

Common pitfalls

Pitfall	Better reasoning
Assuming flatus excludes SBO	Residual distal gas can pass early; complete obstruction can still be present.
Treating all postoperative distention as ileus	Look for transition point, localized pain, hernia, fever, leukocytosis, or acidosis.

Pitfall	Better reasoning
Starting gluten-free diet before testing	Serology and biopsy can normalize; test while the patient is consuming gluten when possible.
Calling chronic diarrhea IBS without red-flag review	Blood, nocturnal stool, anemia, weight loss, hypoalbuminemia, and high calprotectin require evaluation.
Missing medication causes	Metformin, magnesium, antibiotics, PPIs, NSAIDs, SSRIs, colchicine, mycophenolate, and laxatives are common culprits.
Equating all steatorrhea with pancreatic disease	Mucosal disease, bile acid deficiency, SIBO, and lymphatic obstruction can also cause fatty stools.

High-yield numeric anchors

Anchor	Meaning
>4 weeks	Common threshold for chronic diarrhea
Stool osmotic gap <50 mOsm/kg	Supports secretory diarrhea
Stool osmotic gap >100 mOsm/kg	Supports osmotic diarrhea
tTG-IgA + total IgA	Common initial celiac testing strategy while patient eats gluten
First-degree relative risk around 10%	Useful celiac screening anchor from the source packet
Dermatitis herpetiformis in 10-20%	Celiac association highlighted in the source packet
SBO hospital stay around 8 days; mortality around 3%	Morbidity anchor from Bologna adhesive SBO guideline
IBD onset often 15-35 years	Epidemiology anchor from the source packet; older onset can still occur
24/7 emergency logic	Peritonitis, ischemia, free air, sepsis, or closed-loop obstruction overrides slow diagnostic workups

Practice questions

Question 1. A 64-year-old man with prior colectomy has colicky abdominal pain, vomiting, distention, and no stool. CT shows dilated proximal small bowel with a transition point and mesenteric swirl. What is the management priority?

Answer. Resuscitate, keep NPO, decompress as needed, start antibiotics if ischemia suspected, and obtain urgent surgical evaluation. The mesenteric swirl/closed-loop concern makes prolonged observation unsafe.

Question 2. A 28-year-old woman has chronic bloating, iron deficiency anemia, and diarrhea. She stopped gluten last month and feels slightly better. Her tTG-IgA is negative. What is the pitfall?

Answer. Testing after gluten withdrawal can be falsely negative. Celiac testing is best performed while the patient is eating gluten; total IgA and appropriate IgG-based tests matter if IgA deficient.

Question 3. A hospitalized older adult on opioids after hip surgery has abdominal distention. Imaging shows diffuse gas in small bowel and colon without transition point. What is the likely diagnosis?

Answer. Paralytic ileus. Treat supportively by correcting electrolytes, reducing opioids/anticholinergics, mobilizing, and decompressing only if needed.

Question 4. A patient has watery diarrhea that persists during fasting and wakes him at night. Stool osmotic gap is low. What category is most likely?

Answer. Secretory diarrhea. Consider bile acid diarrhea, microscopic colitis, endocrine causes, medication effects, and some infections.

Question 5. A patient with systemic sclerosis has bloating, diarrhea, B12 deficiency, and elevated folate. What mechanism links these findings?

Answer. Small intestinal bacterial overgrowth from dysmotility. Bacteria can consume B12 and produce folate while causing gas, diarrhea, and malabsorption.

Graph-RAG concept extraction map

Node	Important edges/relationships
Small bowel obstruction	caused by adhesions/hernias/Crohn/malignancy; causes vomiting, third spacing, hypovolemia; complicated by ischemia/perforation; diagnosed by CT; managed with NPO/IV fluids/NG/surgery

Node	Important edges/relationships
Closed-loop obstruction	is subtype of SBO; increases pressure rapidly; strongly associated with ischemia and emergent surgery
Paralytic ileus	mimics obstruction; caused by surgery/opioids/hypokalemia/peritonitis; shows diffuse gas; treated by correcting cause
Celiac disease	immune reaction to gluten; causes villous atrophy; associated with tTG-IgA, dermatitis herpetiformis, iron deficiency, osteoporosis; treated with gluten-free diet
SIBO	caused by dysmotility/anatomic stasis; causes bloating/diarrhea/B12 deficiency; evaluated by breath testing; treated with antibiotics and cause correction
Chronic diarrhea	classified as watery/fatty/inflammatory; linked to stool osmotic gap, fecal calprotectin, fecal fat, medication review
Malabsorption	causes weight loss and nutrient deficits; localized by iron/folate/B12/ADEK patterns; overlaps with celiac, pancreatic insufficiency, bile acid disease, SIBO

Selected references used for medical cross-checking

American College of Gastroenterology. ACG Clinical Guideline update: diagnosis and management of celiac disease. Am J Gastroenterol. 2023.

American Gastroenterological Association. Clinical practice guideline on laboratory evaluation of functional diarrhea and diarrhea-predominant IBS in adults. Gastroenterology. 2019.

World Society of Emergency Surgery. Bologna guidelines for diagnosis and management of adhesive small bowel obstruction: 2017 update. World J Emerg Surg. 2018.

American College of Gastroenterology. Clinical guideline: small intestinal bacterial overgrowth. Am J Gastroenterol. 2020.

Medvale source packet gi2.pdf was used as a coverage checklist for SBO, ileus, celiac sprue, Whipple disease, tropical sprue, and IBD epidemiology signals; no source figures were reproduced.